

Event sequences and networks 1

V. Batagelj

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Event sequences and networks

part 1. relational events

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Outline

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3 Networks and SA
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Current version of slides (October 23, 2026 at 02 : 04): slides PDF
https://github.com/bavla/SDA/



Sequence analysis

Event sequence data encode sequences of events over time.

Traditional data analysis uses single numbers or distinct categories. SDA handles aggregated or variable data that contains internal variation and/or structure.

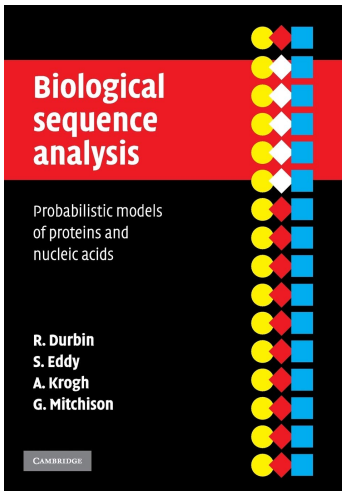
Sequence analysis is a subfield of longitudinal data analysis closely related to time series analysis, but with an emphasis on **categorical** values represented as states. It is described by several tags: sequence analysis, life course, trajectory, career path, event history, multichannel sequence, discrepancy analysis, etc.

There is The Sequence Analysis Association (SAA).

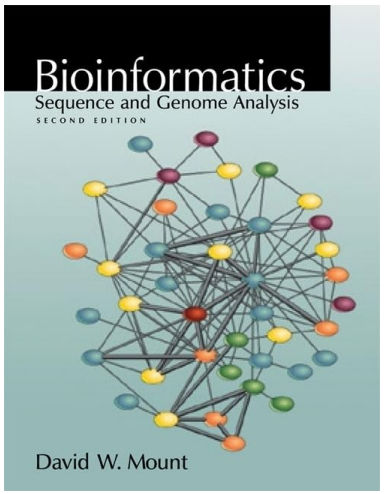
For recent overviews, see Liao et al. [2022], Yeshchenko and Mendling [2023], van der Linden et al. [2023], Raab and Struffolino [2022].

Sequence analysis

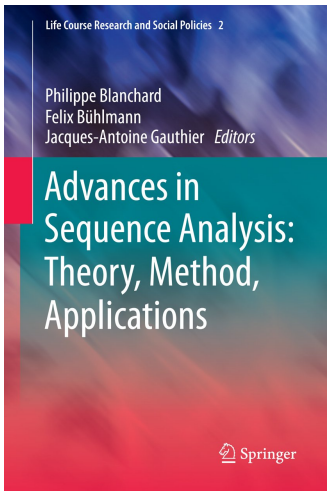
Books



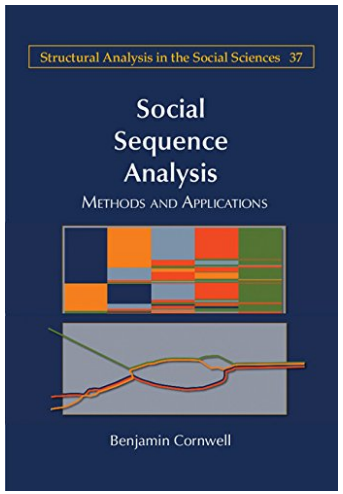
1998



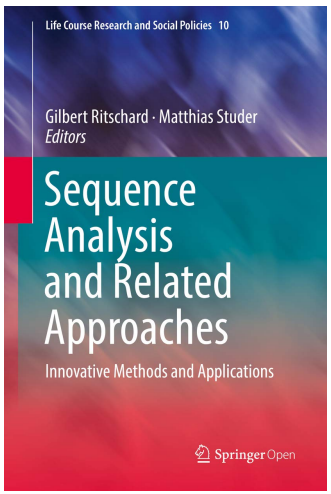
2004



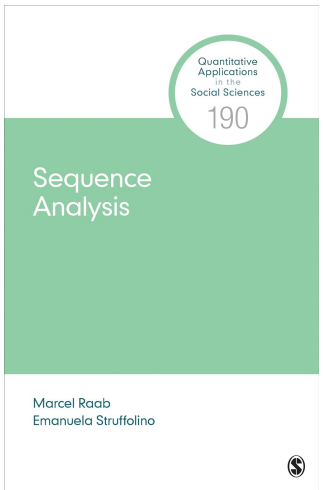
2014



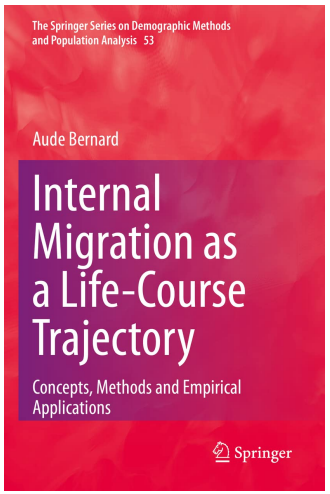
2015



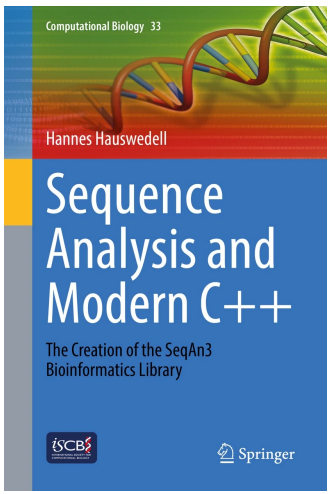
2018



2022



2022



2022

Sequence analysis
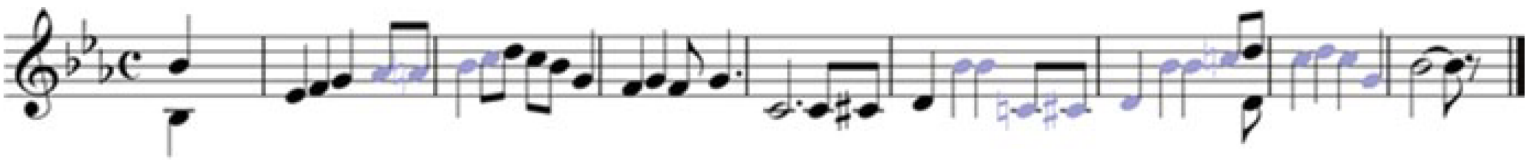
Example gene homologies / Pax6 gene

Mouse	GTATCCAACGGTTGTGTGAGTAAAATTCTGGGCAGGTATTACGAGACTGGCTCCATCAGA
Fly	GTATCAAATGGATGTGTGAGCAAAATCCTCGGGAGGTATTATGAAACAGGAACCATACGA
Shark	GTGTCCAACGGTTGTGTCAGTAAAATCCTGGGCAGATACTATGAAACAGGATCCATCAGA
Squid	GTCTCCAACGGCTGCGTTAGCAAGATTCTCGGCAGGTACTATGAGACGGGCTCCATAAGA
Worm	GTGTCTAATGGTTGTGTTAGTAAAATACTTTGCCGATATTATGGAACAGGTTCTATTAAA
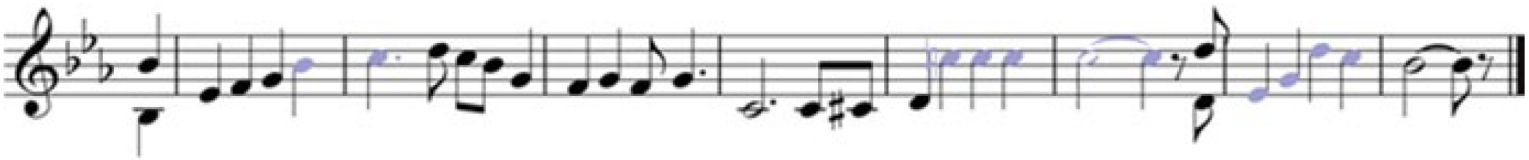
video

Sequence analysis

Example music / hein-v-harris



Silvio Hein, "Maria Cahill's Arab Love Song"



Joseph E. Howard, "I Think I Hear a Woodpecker"

Sequence analysis

Bikes sharing systems / GitHub/bavla/Bikes

Standard CSV Trip Data Fields

Most major systems (such as Citi Bike, Metro Bike Share, and Capital Bikeshare) structure their historical trip records with the following core columns: Metro Bike Share +2

- **Trip Identifiers:** `trip_id` or `ride_id` (a unique alphanumeric string for each ride).
- **Timestamps:** `start_time` / `started_at` and `stop_time` / `ended_at` (recorded in local system time or UTC).
- **Duration:** `trip_duration` (measured in seconds or minutes).
- **Start Station Info:** `start_station_id`, `start_station_name`, along with `start_lat` and `start_lng`.
- **End Station Info:** `end_station_id`, `end_station_name`, along with `end_lat` and `end_lng`.
- **Bike Information:** `bike_id` (unique bike identifier) and `bike_type` (classic vs. electric/e-bike).
- **Rider Profile:** `user_type` / `member_type` (subscriber/annual member vs. casual/single-trip rider), and occasionally anonymized demographic data like birth year or gender. Capital Bikeshare DC +4

Sequence analysis

Example political event data / KEDS, Cameo

```
...
910625  YUG     YUGCRO  198     (WITHDRAW FROM) DECLARED INDEPENDENCE
910625  YUG     SLO     198     (WITHDRAW FROM) DECLARED INDEPENDENCE
910625  YUGCRO  YUG     081     (MAKE AGREEMENT)        IS PLANS
910625  YUGCRO  YUG     198     (WITHDRAW FROM) DECLARE INDEPENDENCE
910625  USA     YUG     102     (URGE)  REITERATED
910625  GER     YUG     031     (MEET)  TELEPHONED
910625  YUG     GER     031     (MEET)  TELEPHONED
910625  ITA     YUG     031     (MEET)  TELEPHONED
910625  YUG     ITA     031     (MEET)  TELEPHONED
910625  FRN     SLO     121     (CRITICIZE)     DISAPPROVED
910625  FRN     YUGCRO  121     (CRITICIZE)     DISAPPROVED
910626  SLO     YUG     043     (RALLY) CELEBRATED
910626  YUGCRO  SERMIL  223     (MIL ENGAGEMENT)        KILLED
910626  USR     YUGCRO  042     (ENDORSE)       SOVIET SUPPORT FOR CROATIA
910626  USR     SLO     042     (ENDORSE)       SOVIET SUPPORT FOR CROATIA
910626  YUG     SLO     182     (MILITARY DEMO) PUT ON ALERT
910626  YUG     YUGCRO  223     (MIL ENGAGEMENT)        KILLED
910626  SERMIL  SLO     199     ()      COMBAT
910626  YUGGOV  SLO     197     (CENSOR)        OUTLAWED
910626  YUGGOV  YUGCRO  197     (CENSOR)        OUTLAWED
910626  YUGGOV  SLO     210     (SEIZE) SEIZE
910626  AUL     YUGCRO  022     (PESSIMIST COMMENT)     SAID NOT
910626  USAGOV  YUG     022     (PESSIMIST COMMENT)     SAID NOT
910626  USA     YUG     112     (REFUSE)        OPPOSED
910626  USA     EUR     160     (WARN)  WARNED
910626  UNK     YUG     094     (CALL FOR)      CALLED ON
910626  NTH     EEC     022     (PESSIMIST COMMENT)     SAID NOT
910626  GERGOV  YUG     102     (URGE)  URGED
910627  SLO     SERMIL  223     (MIL ENGAGEMENT)        FIGHTING ERUPTED
910627  SLO     SERMIL  223     (MIL ENGAGEMENT)        SHOT
910627  SERMIL  SLO     031     (MEET)  SENT
910627  SLO     SERMIL  121     (CRITICIZE)     SLOVENIA SAID YUGOSLAV SOLDIER KILLED
...
```

Sequence analysis

Example purchase



SPAR

SUPERMARKET KAPITELJ, Poljanski nasip 8, 1000 Ljubljana
Spar Slovenija d.o.o., Letališka c. 26, 1000 Ljubljana
ID DDV SI32156782

****

Ljubljana 04.03.2024 15:57 Bl 101 Bon 1442 U 226

Izdelek	Količina	Cena	Popust	Znesek
OLJE OLJČNO ILIADA 1L	1	21.98	5.50	16.48 B
STEGNA PIŠČ.PRE.BKK IK	1	5.69	0.00	5.69 B
BLAZINICE NIVEA 80/1	1	2.59	0.00	2.59 C
MASLO PLANIKA 250G	1	3.89	0.00	3.89 B
PAPIR TOAL.ZEWA 10/1	1	5.99	2.00	3.99 C
FIŽOL RJAVI 190G	2	1.15	0.00	2.30 B
SPAR PR. MUSLI MAND.37	1	3.79	0.00	3.79 B
VINO MALVAZ.VK 0,75L	1	4.99	0.00	4.99 C
BANANE	1.020	1.49	0.61	0.91 B
JAJCA M 6/1	1	2.68	0.00	2.68 B
Skupaj		**EUR**		**47.31**

Sequence analysis

Example RIS (Research Information Systems)

```
PT  - J
AU  - Watson, James D.
      Crick, Francis H. C.
TI  - Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid
SO  - Nature
PY  - 1953
VL  - 171
IS  - 4356
BP  - 737
EP  - 738
CR  - Astbury WT, 1947, SYMP SOC EXP BIOL, V1, P66
      Avery OT, 1944, J EXP MED, V79, P137
      Chargaff E, 1950, EXPERIENTIA, V6, P201
      Wilkins MH, 1953, NATURE, V171, P738
ER  -
```

Raw event sequence

Usually we analyze a finite set **S** of event sequences.

A *raw event sequence* S is a list of events $S = (e_1, e_2, \ldots, e_n)$, n is the *length* of the sequence S. Each event is a list of *named attribute values*

$$e_i = (ind = i, t = t_i, a_1 = a_{1i}, a_2 = a_{2i}, \ldots, a_m = a_{mi})$$

where *ind* is the event's *index*, t is the event's *temporal* information, and a_is are *data attributes* describing different event characteristics. Their values can be structured/symbolic objects.

In some events, attributes can be missing. Each event has at least one data attribute defined (has a value). An attribute a_i is called *general* if it is defined for all events in the sequence S. The attribute *ind* and, usually, the time t are general attributes.

From S we usually obtain the set $\mathbf{S} = \{S_x\}$ by partitioning S by selected categorical attribute a: $S_x = \{e_i \in S : a_i = x\}$.

Event attributes / Time

The temporal information can be described in different ways

- implicitly – the event's index *ind* determines the time. It expresses the ordering of events,
- $t_i \in \mathbb{N}$ – it expresses the ordering of events or starts of events on a discrete timeline (hours, days, years, etc.),
- $t_i \in \mathbb{R}$ – time instant
- $t_i = (s_i, f_i)$ – it specifies the time interval of event e_i; s_i is the start and f_i is the finish time of the event; $s_i \leq f_i$,
- $t_i = (s_i, d_i)$ – it specifies the time interval of event e_i; s_i is the start and d_i is the duration time of the event; $d_i \geq 0$.

. . . Event attributes

Quantities s_i and f_i are expressed as numbers or as date-time stamps. In some applications, an event's time can be of different types: *interval* (both values are defined) and *instantaneous* (the second value is undefined). The time can be measured on a *common* timeline or a *relative* timeline (for example, from each person's birthday).

Often in our analyses, we do not need all available data attributes, and some attributes need to be recoded – we transform the raw event sequence S into a *work* event sequence S'.

Usually, we also have some additional data about the (values of) attributes used in the event description. For example: address, age, gender, etc. of persons. Several event sequence datasets are available [Batagelj, 2026].

Questions

Sequence analysis (SA) deals with questions about the set of event sequences **S**

- structure of the set **S** (statistics, clustering, visualization)
- models that generate sequences from the set **S** (transition matrices, Petri nets, Markov chains)
- frequent patterns – motifs (subsequences) in sequences from **S**
- networks derived from **S**
- etc.

Derived data sets

Attribute values in events can be numerical, categorical (nominal) or structured.

Categorical attributes (CA) can be used to construct various derived data sets:

- the values of a selected CA are summarized by their distribution
- a selected CA defines states – partitioning the data into a set of event sequences (standard SA)
- two selected CAs define sets of nodes in a (2-mode) network
- k selected CAs define sets of nodes in a k-way network
- a selected CA and an attribute whose values are subsets define a 2-mode network or a hypergraph
- ...

Networks and SA

In this presentation, we will focus on using networks to analyze sets of event sequences. In some cases, events are *relational* – two general CAs define the nodes of the network.

For example, in bike rental events, there is a rental/start station and a return station. Aggregated values of parallel links can be described by symbolic objects.

Sometimes, however, we can take sequences of events from **S** as nodes. Nodes are *linked* if the sequences are active at least for some time simultaneously [Oštro, 2024] → Part 2.

Time

Selecting an appropriate granulation (5 min, 15 min, 30 min, 1 hour, part of a day, day, week, day of the week, month, quarter, year, etc.) and some restrictions (rider type, gender, age, ...), we get the corresponding frequency distributions in nodes and on links.

Transformation of original times Y/M/D/h/m/s

Monthly activity: (t, M)

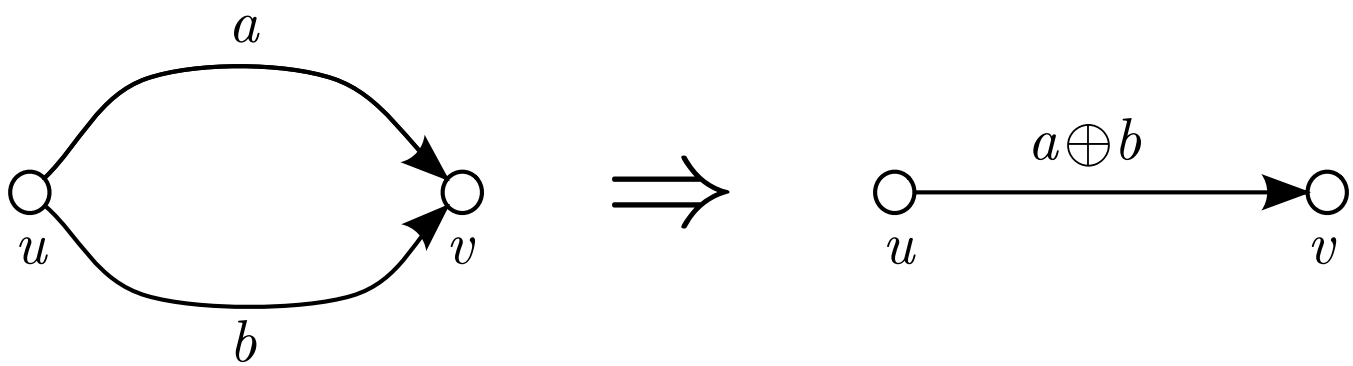
Year-by-month activity: $(t, Y/M)$

Day in a year activity: (t, dy)

Day of the week by hour activity: $(t, dw/h)$ Wikipedia [2026]

Networks and semirings

We want to extend a weight *w* in a network to subsets of links, walks, and sets of walks. We need two operations on weights



Semirings are the right algebraic structure for this.

Temporal quantities

We introduce a notion of a *temporal quantity* (TQ) [Batagelj and Praprotnik, 2016]

$$a(t) = \begin{cases} a'(t) & t \in T_a \\ \text{⌘} & t \in \mathcal{T} \setminus T_a \end{cases}$$

where T_a is the *activity time set* of a and $a'(t)$ is the value of a in an instant $t \in T_a$, and ⌘ denotes the value *undefined*.

We assume that the values of TQs belong to a set A which is a *semiring* $(A, +, \cdot, 0, 1)$ for binary operations $+$ and $\cdot$.

We can extend both operations to the set $A_{\text{⌘}} = A \cup \{\text{⌘}\}$ by requiring that for all $a \in A_{\text{⌘}}$ it holds

$$a + \text{⌘} = \text{⌘} + a = a \quad \text{and} \quad a \cdot \text{⌘} = \text{⌘} \cdot a = \text{⌘}.$$

The structure $(A_{\text{⌘}}, +, \cdot, \text{⌘}, 1)$ is also a semiring.

Operations with temporal quantities

Let $A_{\Cmd}(\mathcal{T})$ denote the set of all TQs over $A_{\Cmd}$ in time $\mathcal{T}$. To extend the operations to networks and their matrices we first define the *sum* (parallel links) $a \oplus b$ as

$$(a \oplus b)(t) = a(t) + b(t) \quad \text{and} \quad T_{a \oplus b} = T_a \cup T_b.$$

The *product* (sequential links) $a \odot b$ is defined as

$$(a \odot b)(t) = a(t) \cdot b(t) \quad \text{and} \quad T_{a \odot b} = T_a \cap T_b.$$

Let us define the TQs $\mathbf{0}$ and $\mathbf{1}$ with requirements $\mathbf{0}(t) = \Cmd$ and $\mathbf{1}(t) = 1$ for all $t \in \mathcal{T}$. Again, the structure $(A_{\Cmd}(\mathcal{T}), \oplus, \odot, \mathbf{0}, \mathbf{1})$ is a semiring.

Temporal quantities

To produce software support for computation with TQs, we limit it to TQs that can be described as a sequence of disjoint time intervals with a constant value

$$a = [(s_i, f_i, v_i)]_{i \in 1..k}$$

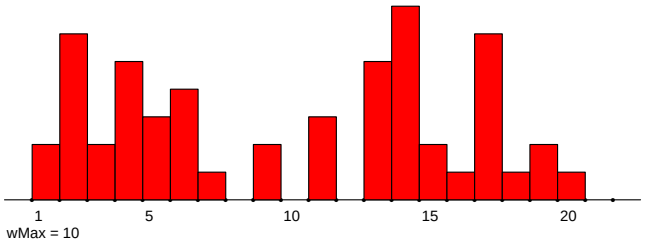
where s_i is the starting time and f_i the finishing time of the i-th time interval $[s_i, f_i)$, $s_i < f_i$ and $f_i \leq s_{i+1}$, and v_i is the value of a on this interval. Outside the intervals, the value of TQ a is undedined, ⌘.

To illustrate both operations let us consider temporal quantities a and b [Batagelj and Praprotnik, 2016]:

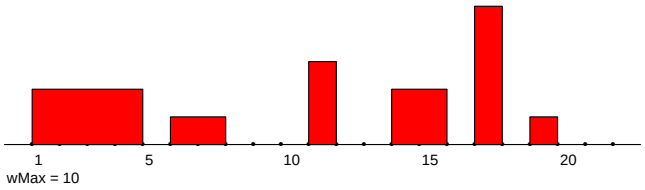
$$\begin{aligned} a &= [(1,5,2),(6,8,1),(11,12,3),(14,16,2),(17,18,5),(19,20,1)] \\ b &= [(2,3,4),(4,7,3),(9,10,2),(13,15,5),(16,21,1)] \end{aligned}$$

Addition and multiplication of TQs

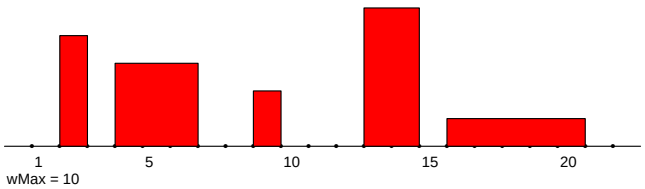
$a \oplus b$:



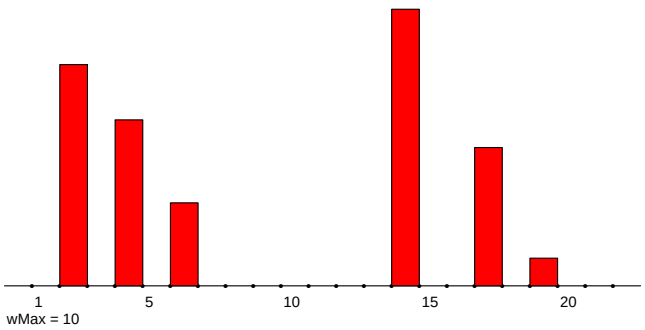
a :



b :



$a \odot b$:



Symbolic networks

Selecting an appropriate granulation, we get the corresponding frequency distributions in nodes and on links. Assigning distributions to nodes and links we get a *symbolic network*.

There are different distributions on event sequence S for time interval t:
starts(S,t) = # of events from S starting in t,
stops(S,t) = # of events from S stoping in t,
active(S,t) = # of events from S active in t (weak, strict),
duration(S,t) = # of events from S with duration in t,
etc.

For example, in a bike-sharing network the event sequence $S(u,v)$ = set of trips starting in station u and finishing in station v.

flow$(u,v;t)$ = active$(S(u,v),t)$ = # of trips starting in station u and finishing in station v weakly active in t

Clustering of flows / 7 clusters

NY Citi Bike, Oct 2015 – Sept 2016. 13 266 296 trips, 678 stations

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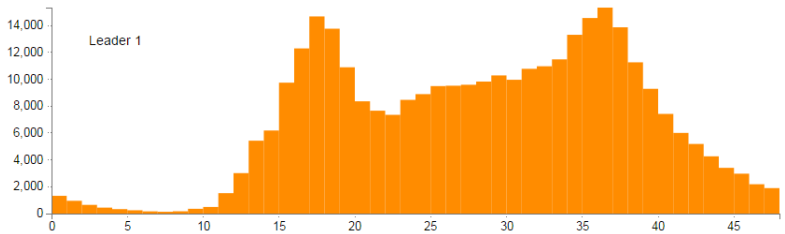
Sequence analysis

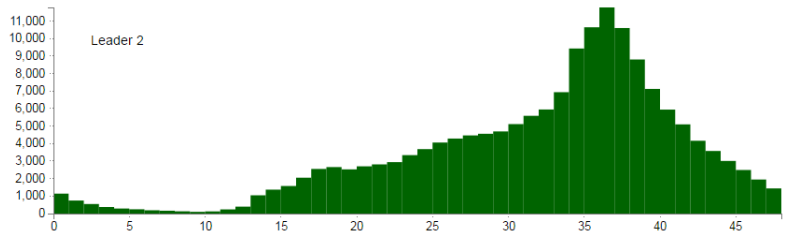
Basic notions

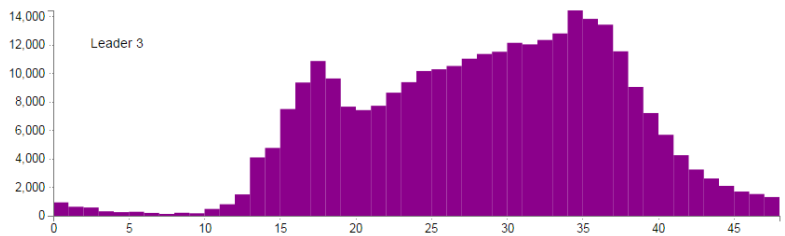
Networks and SA

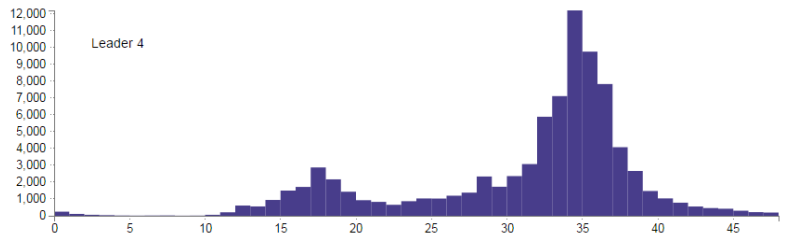
Examples

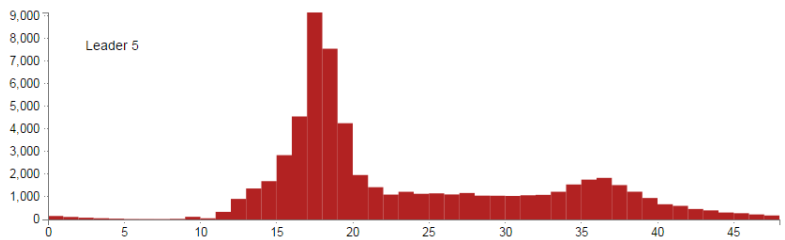
References

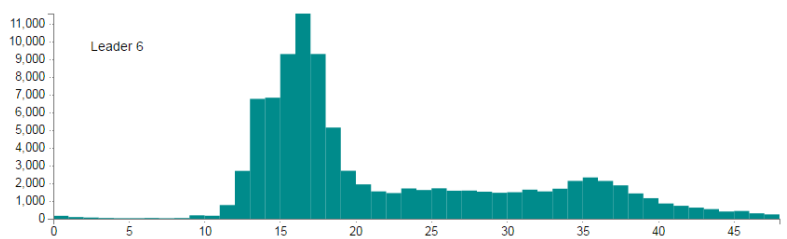


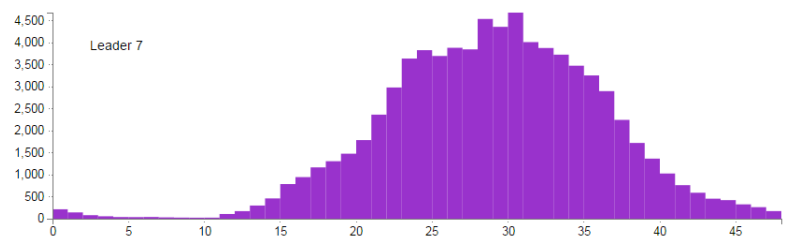




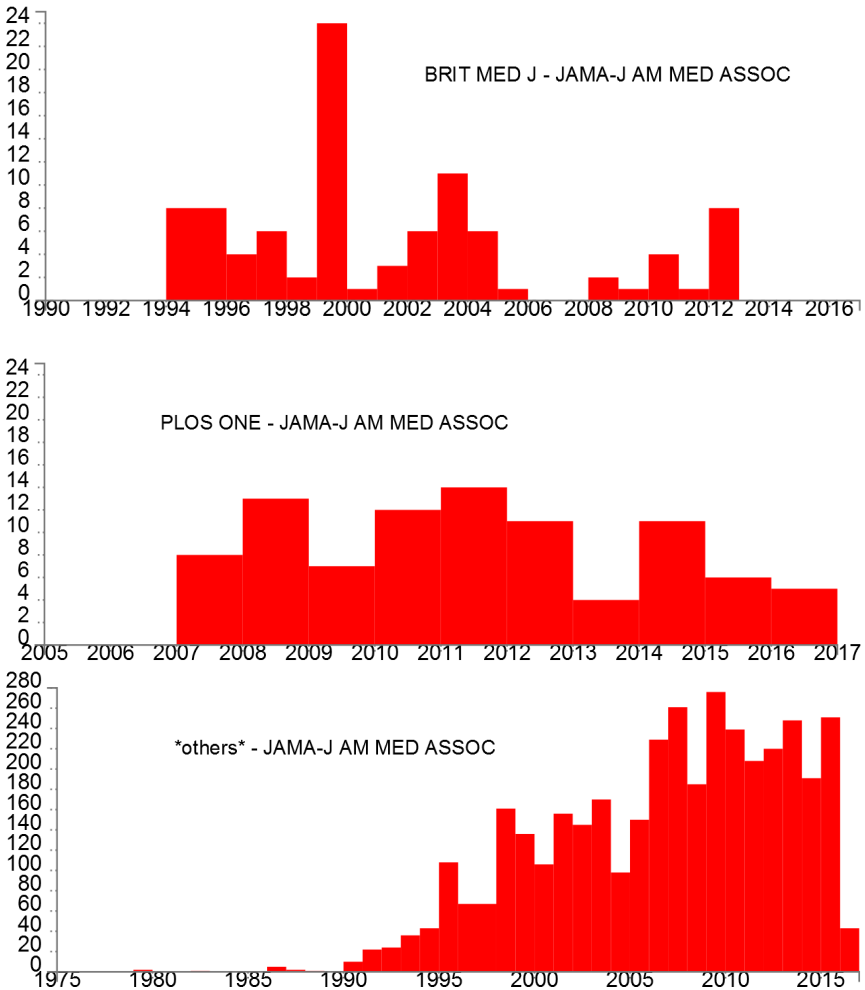








Citations between journals



In the analysis of bibliographic networks, we usually start from basic networks such as **WA**, **Ci**, **WJ**, **WK**, etc. The journal-to-journal citation network, **JCiJ**, is obtained through the multiplication $\mathbf{JCiJ} = \mathbf{WJ}^T \cdot \mathbf{Ci} \cdot \mathbf{WJ}$. This can be generalized to temporal networks [Batagelj and Maltseva, 2020].

Applications

Kejžar et al. [2011], Batagelj and Praprotnik [2016], Batagelj and Maltseva [2020], Batagelj et al. [2022], Batagelj [2024, 2023], Maltseva and Batagelj [2024]



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